universität innsbruck

Juni 30 2025

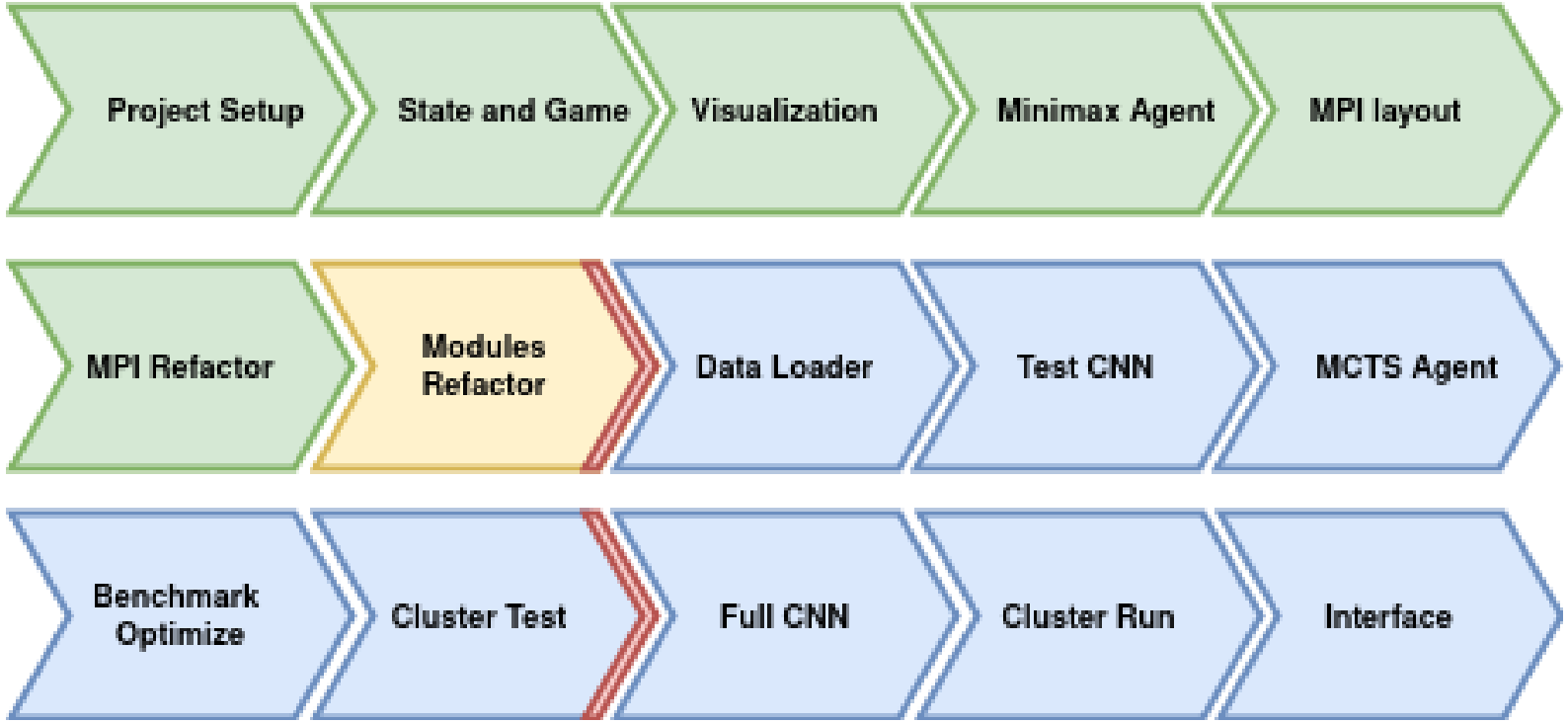
Christof Stein



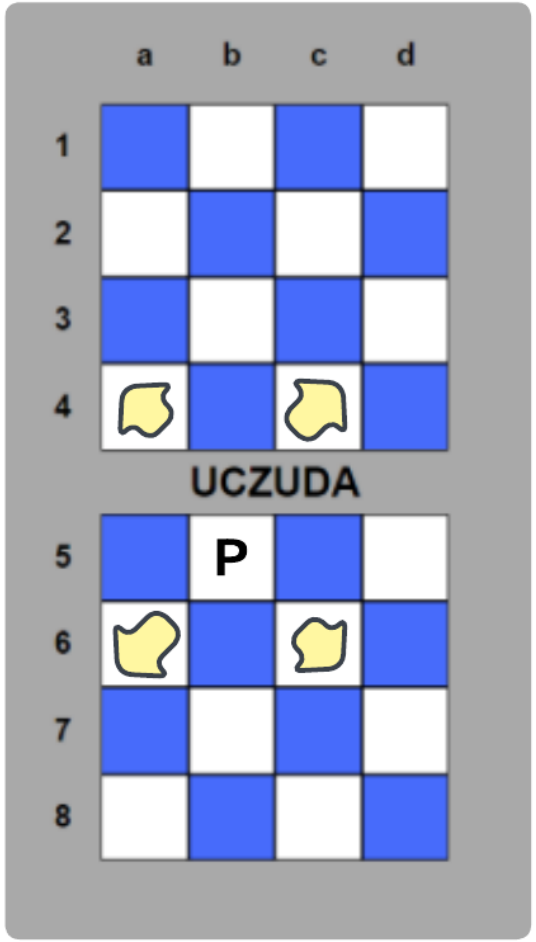
Reinforcement Learing for Zilatran Chess III

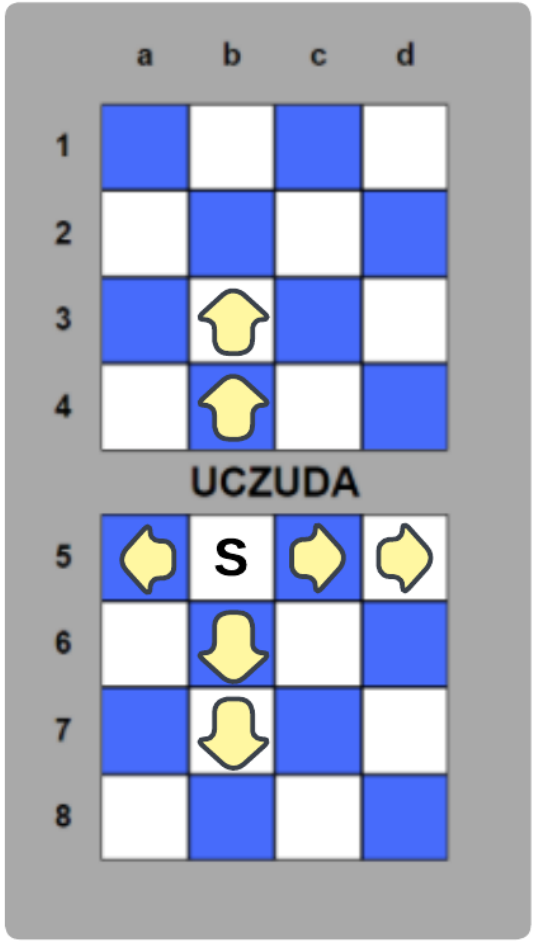
Applying Networks, Loaders to Monte Carlo Tree Search and Deploying to Slurm Clusters

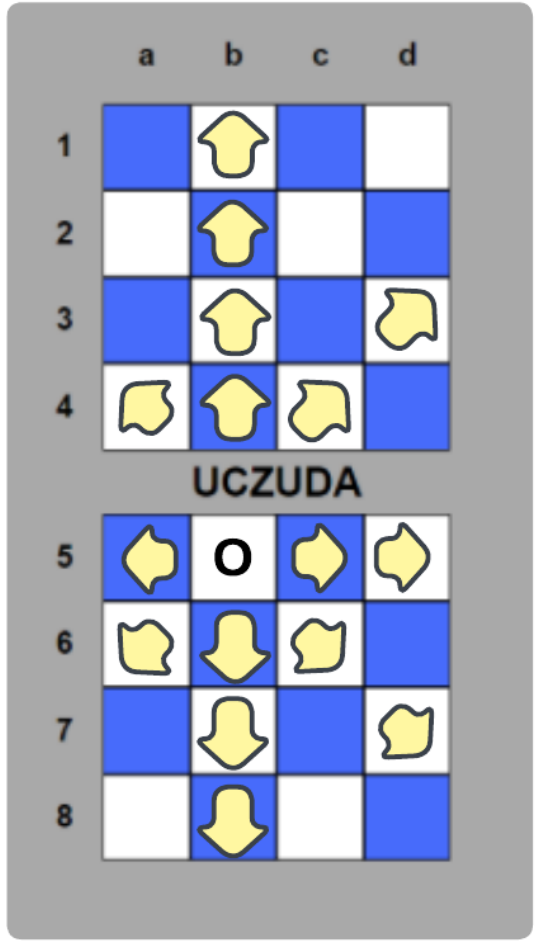
Previously I



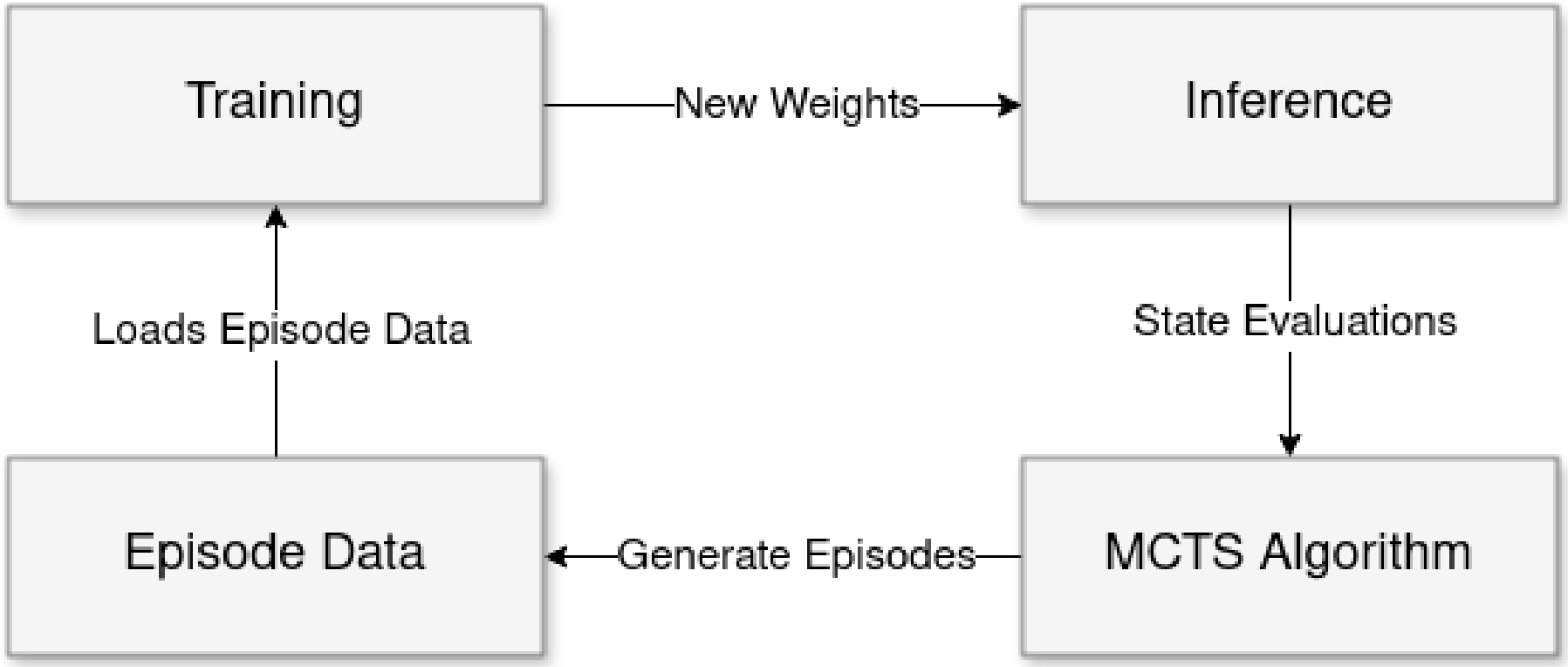
Previously II



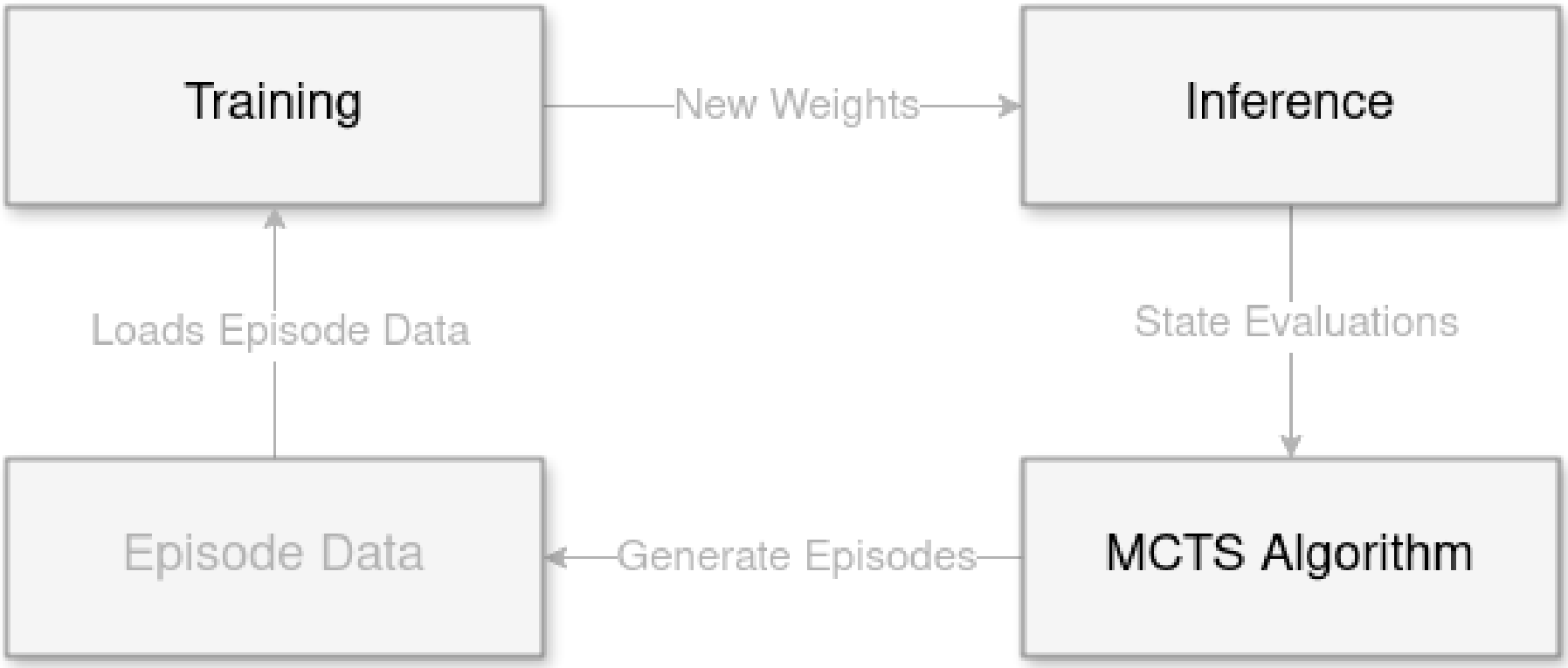




Previously III



What's Missing I

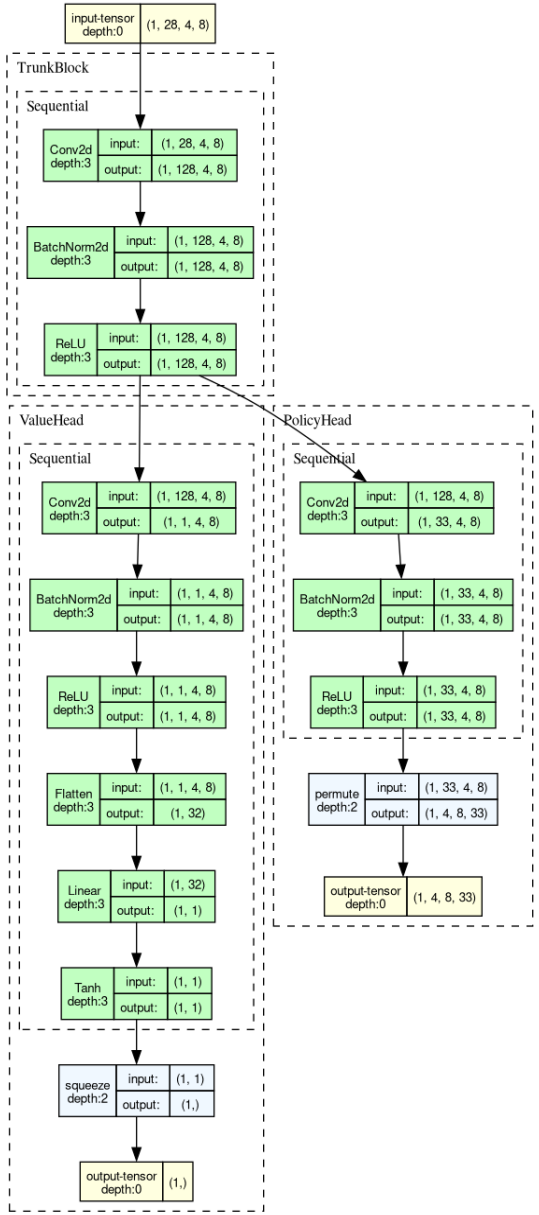


CNNs with pytorch

- Structure
- Loss Function
- Optimizer

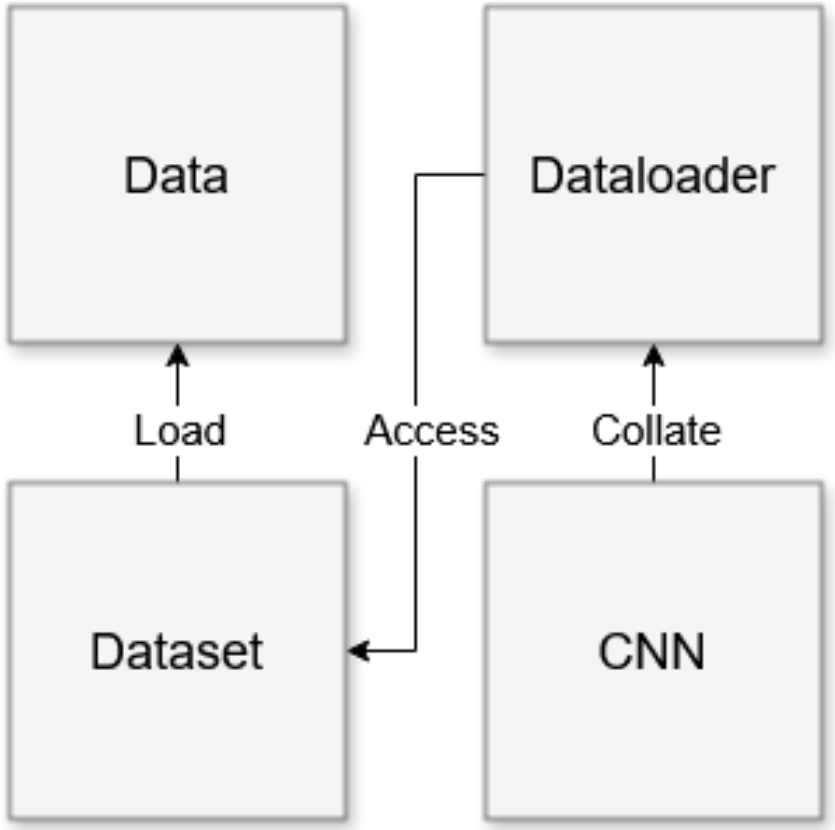
$$\mathcal{L} = (z - v)^2 - \pi^\top \log p + c\|\theta\|^2$$

z is the target outcome (`win/loss/draw`)
v is the value predicted by the network
π is the improved policy from MCTS
p is the predicted policy from the network
θ are the network parameters
c is the L2 regularization coefficient

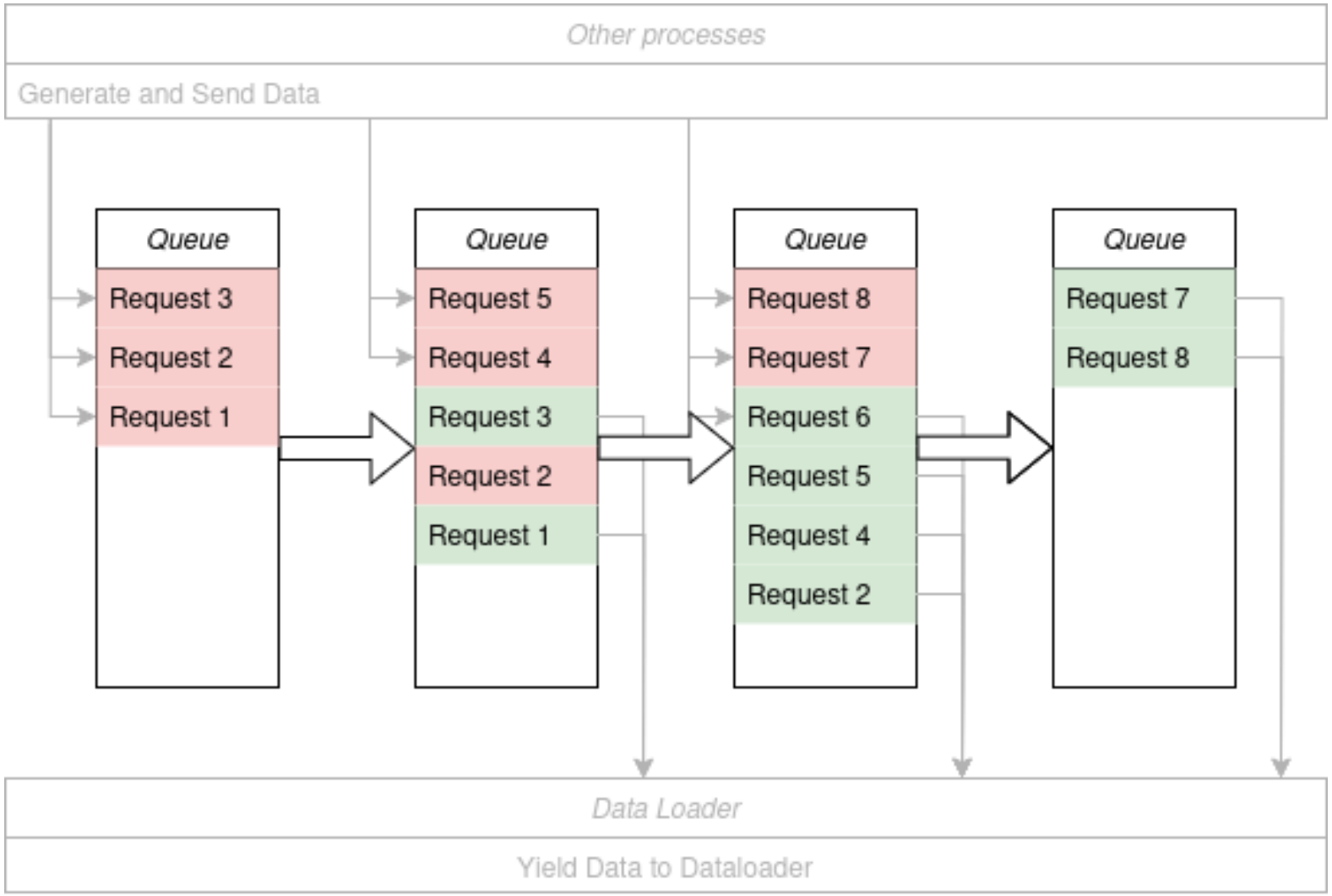


Loading Data I: Concepts

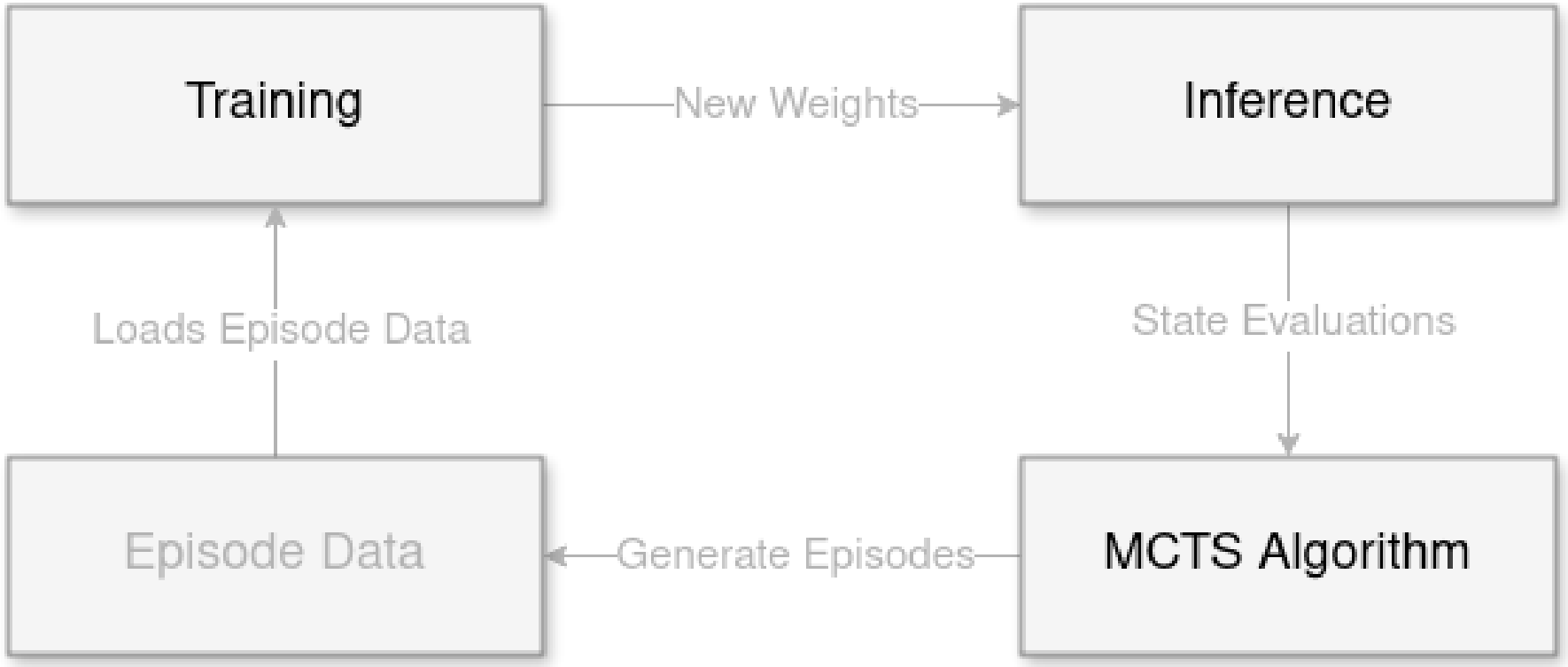
- Datasets and Dataloaders:
  - Lazy Loading
  - Parallel Loading
  - Encapsulate Data Gathering/Batching
  - Industry Standard/Integration



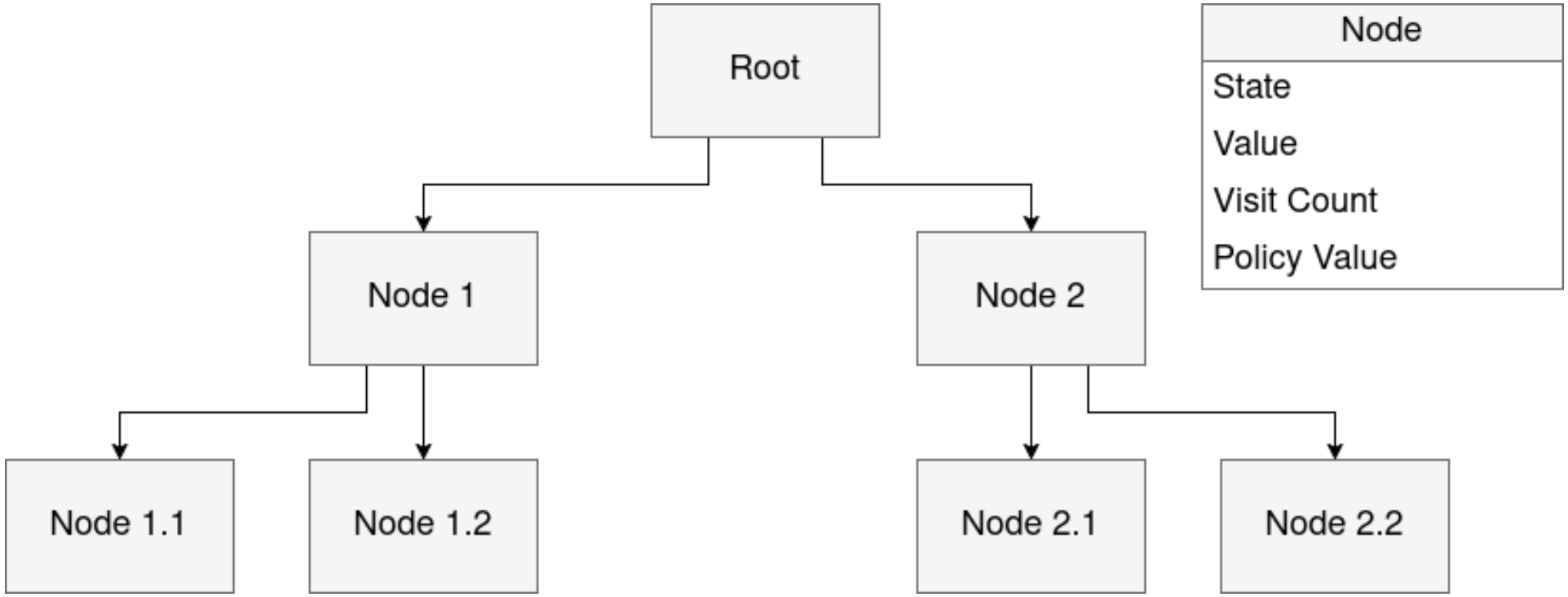
Loading Data II: Dataset with Futures



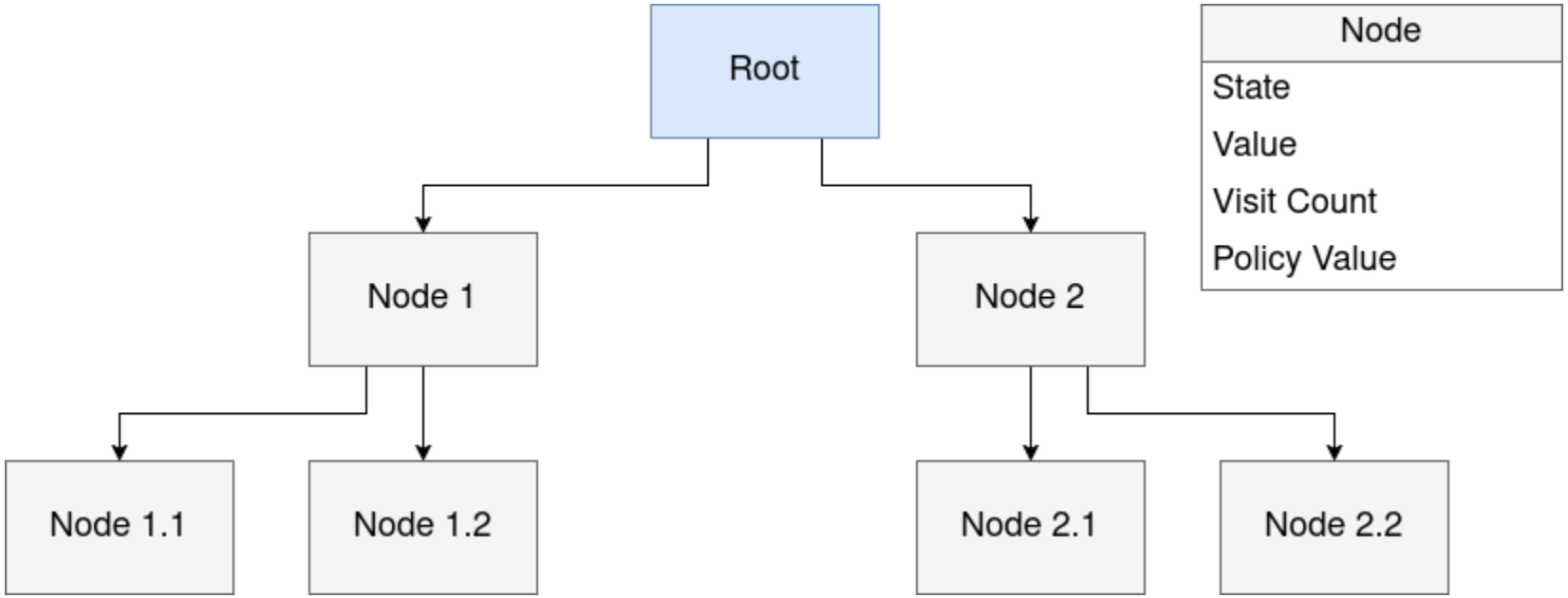
What's Missing II



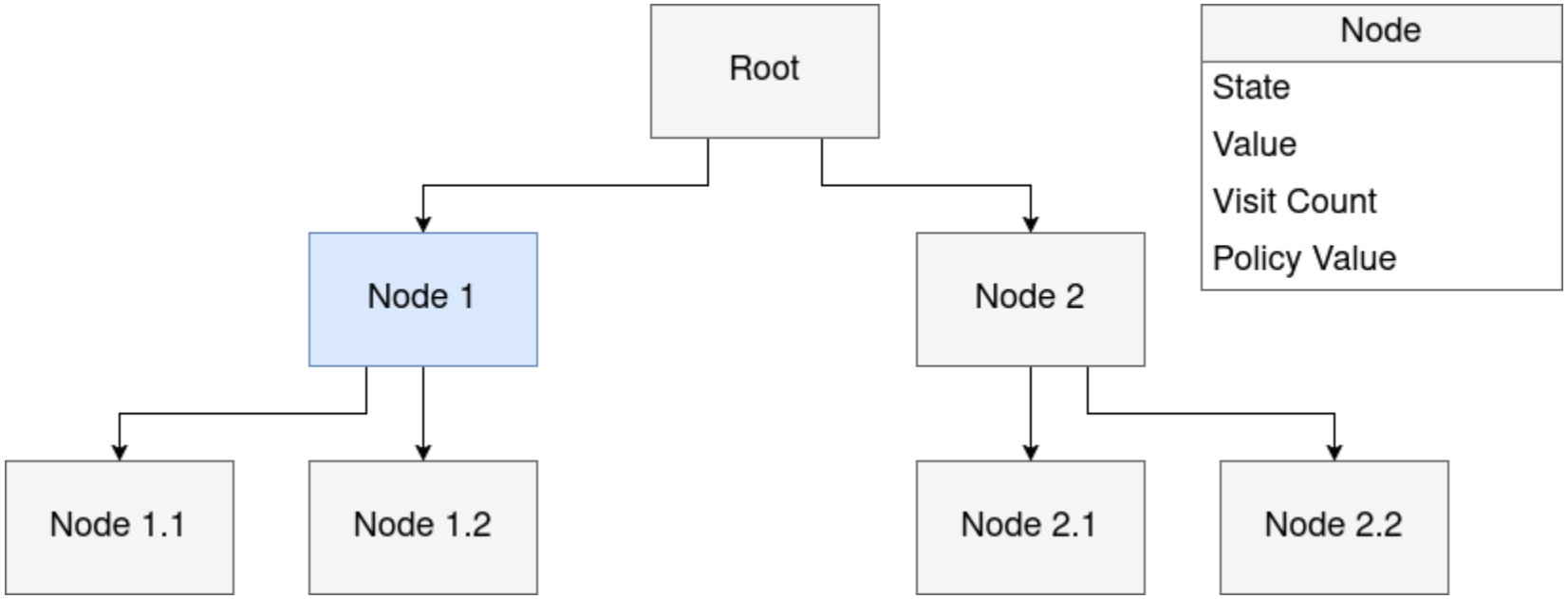
MCTS Algorithm I: Trees and Nodes



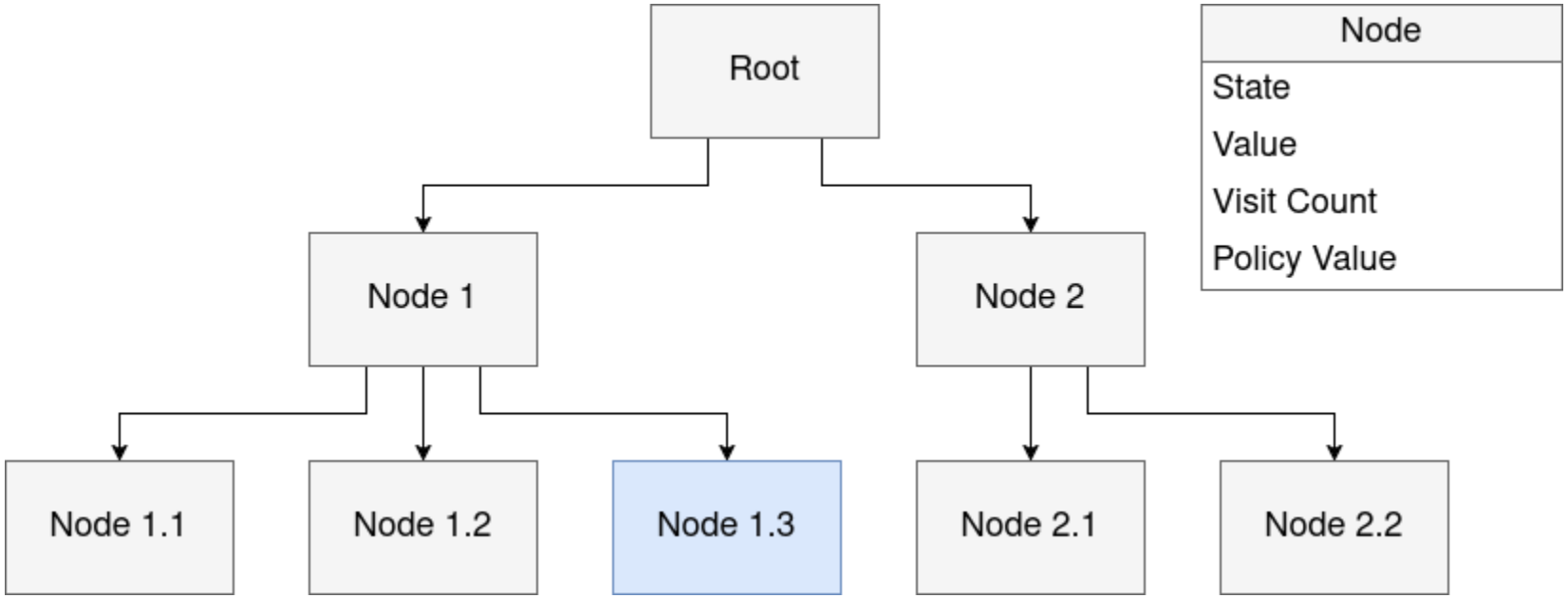
MCTS Algorithm II: Rollout



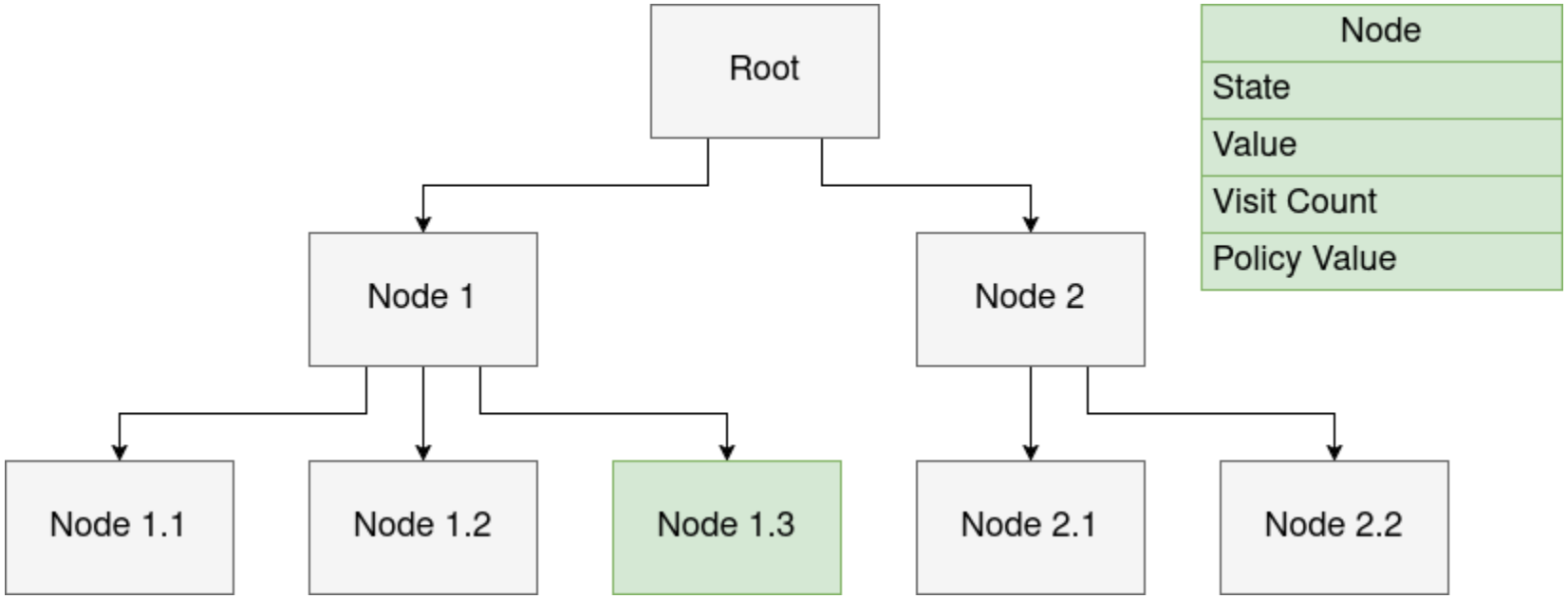
MCTS Algorithm II: Rollout



MCTS Algorithm II: Rollout



MCTS Algorithm III: Backpropagation

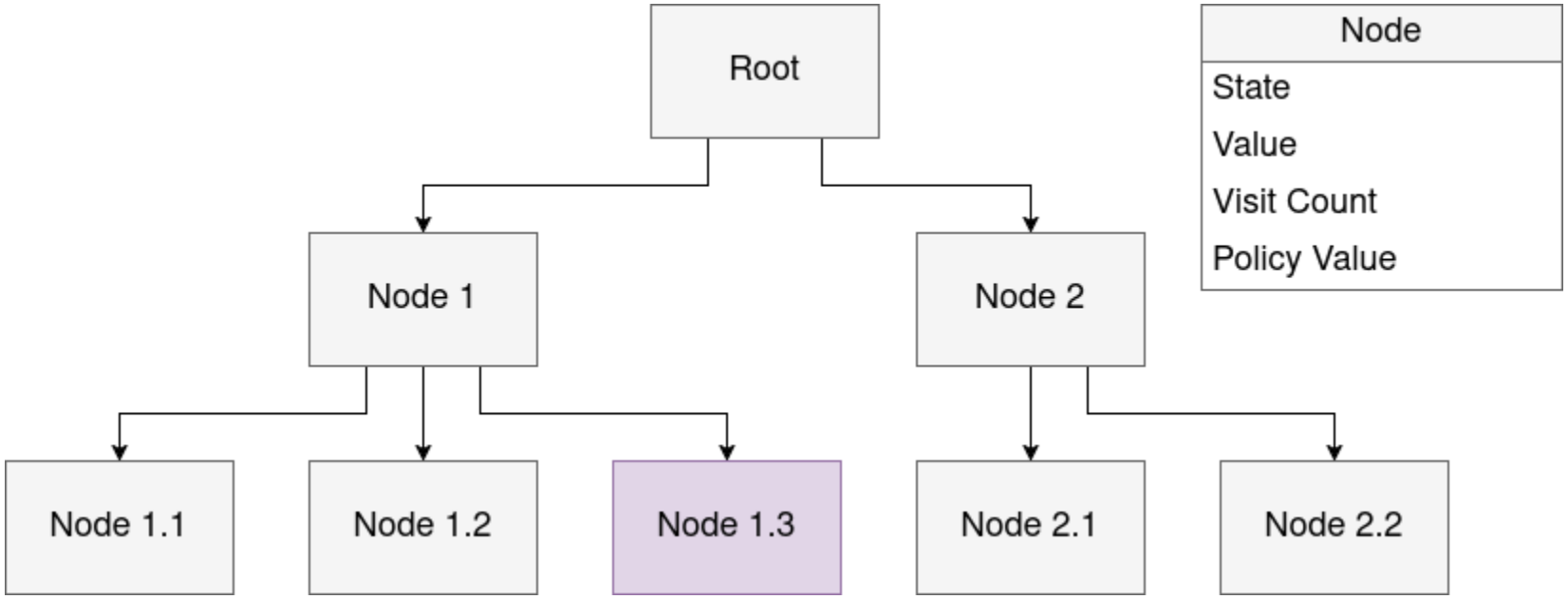


MCTS Algorithm III: Backpropagation

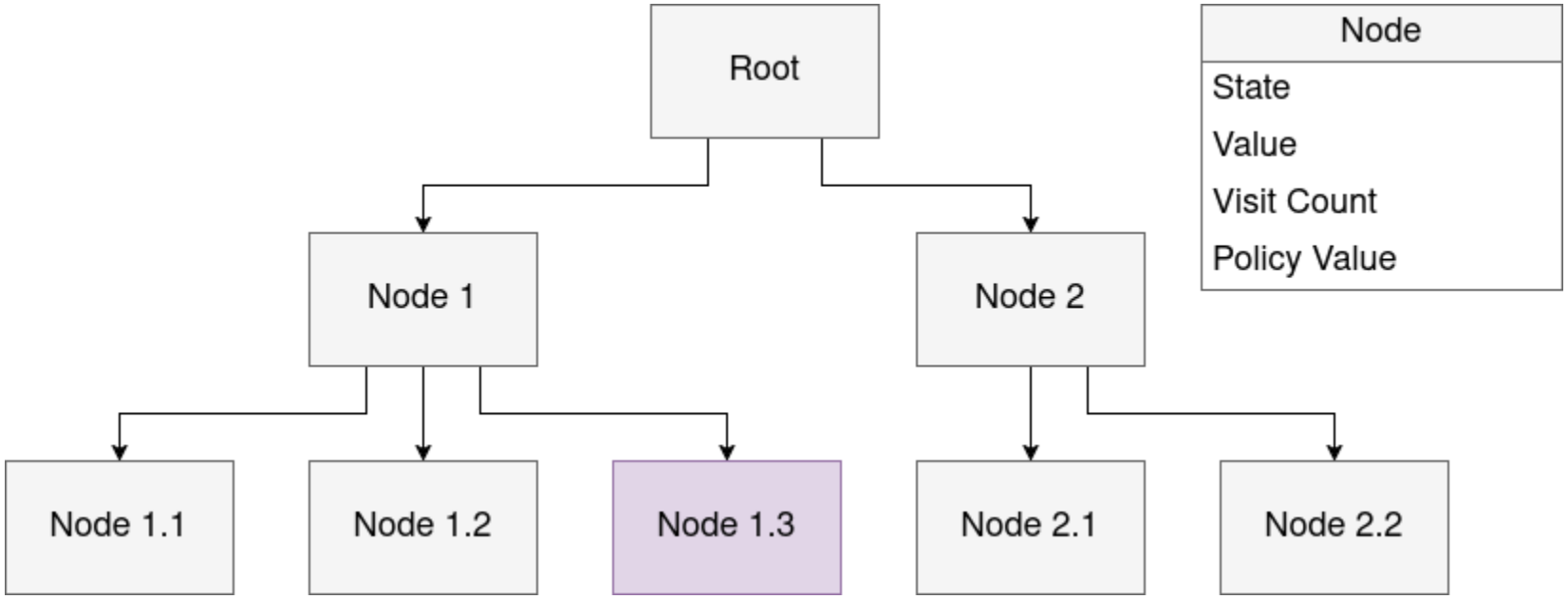
$$PUCT(a) = Q(s,a) + c_{puct} \cdot P(s,a) \cdot \frac{\sqrt{N(s)}}{1 + N(s,a)}$$

$Q(s,a)$: estimated value for action a in state s
$P(s,a)$: prior probability from the policy network
$N(s)$: total number of visits to state s
$N(s,a)$: number of times action a was taken from s
c_{puct}: exploration constant

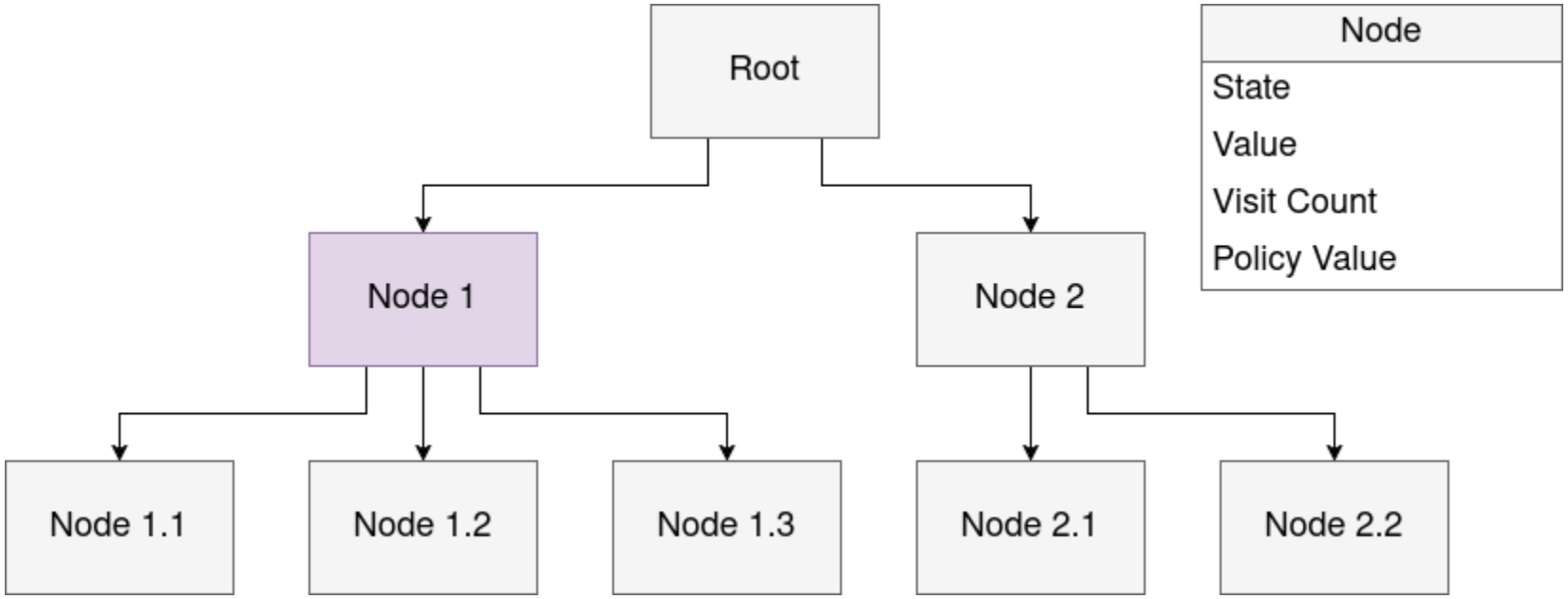
MCTS Algorithm III: Backpropagation



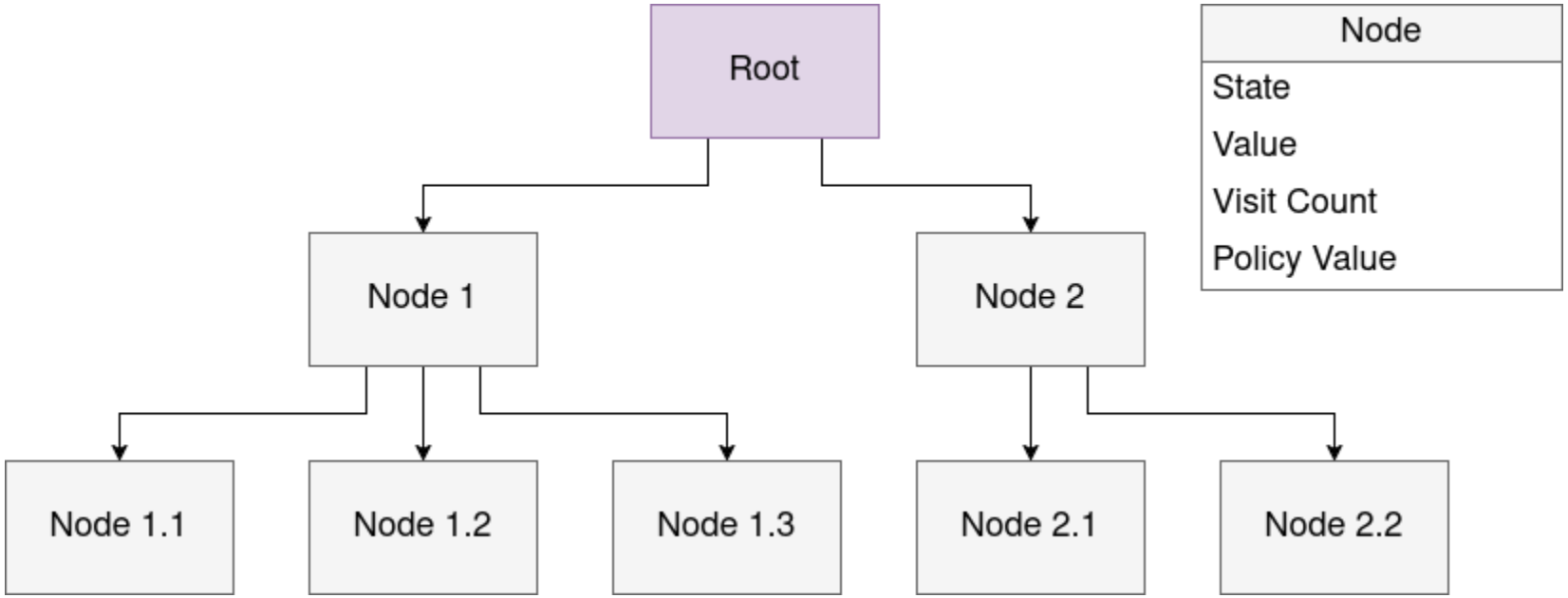
MCTS Algorithm III: Backpropagation



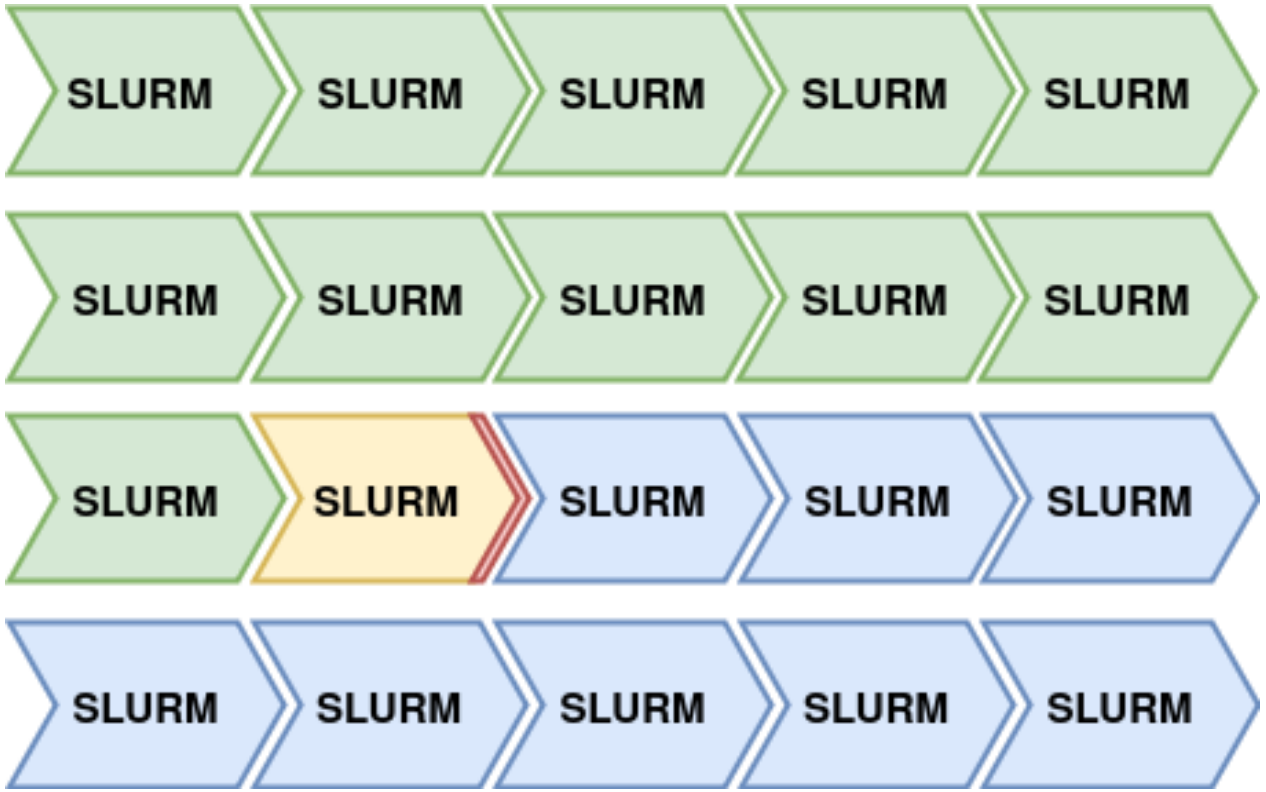
MCTS Algorithm III: Backpropagation



MCTS Algorithm III: Backpropagation



Slurm Setup I



Slurm Setup II

	Hom. Jobs	Het. Job single	Het. Job multi	Srun Config	Reservation
Resource Locality	✗	✓	✓	?	✓
COMM_WORLD	✓	✓	✗ (Sockets)	✓	✓
CPU Resources	✗	✓	✓	✗	?
GPU Resources	✗	✓	✓	✗ (Shared)	?
Notable Issues	GPU mapping	PMI2 Error	Sockets setup	Socket GPU	?
Ease of Use	Simple	Script Only	Script Only	Script Only	Reservation
Working	✗	Small Jobs	✗	Slow	?

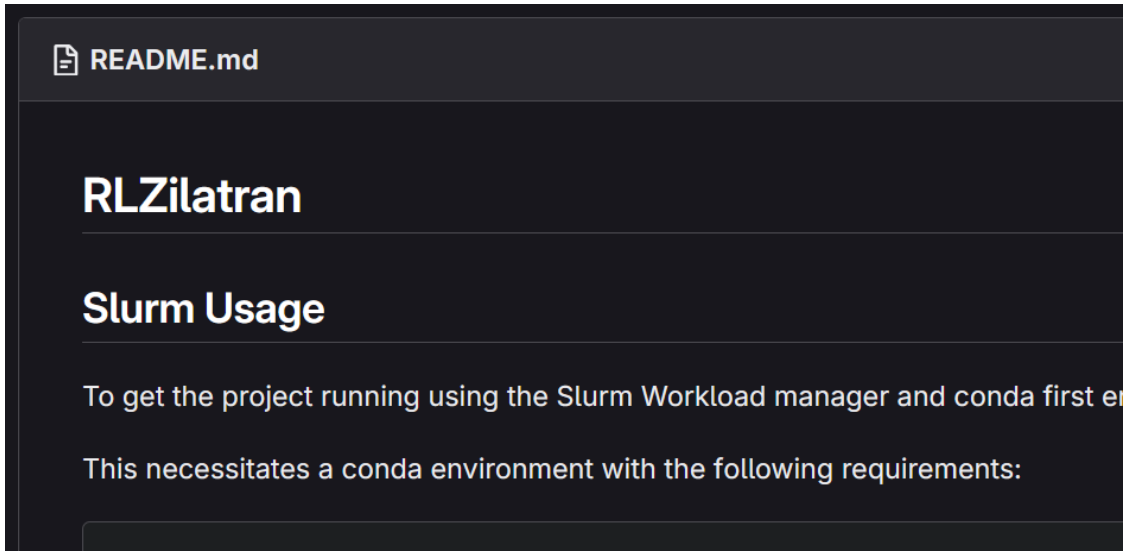
Outlook

- Hetjobs (Again)
- New execution Strategy
- VSC5 Access (Software Stack)
- Improve Tests/Documentation/Proposal

- Hackathon (October)

```
# GPU process block
#SBATCH --ntasks={{ gpu_processes }}
#SBATCH --gres=gpu:a30:{{ gpu_processes }}
#SBATCH --cpus-per-task={{ gpu_threads }}
#SBATCH --mem={{ gpu_memory }}MB

# CPU process block
#SBATCH hetjob
#SBATCH --ntasks={{ cpu_processes }}
#SBATCH --cpus-per-task={{ cpu_threads }}
#SBATCH --mem={{ cpu_memory }}MB
```


README.md

RLZilatran

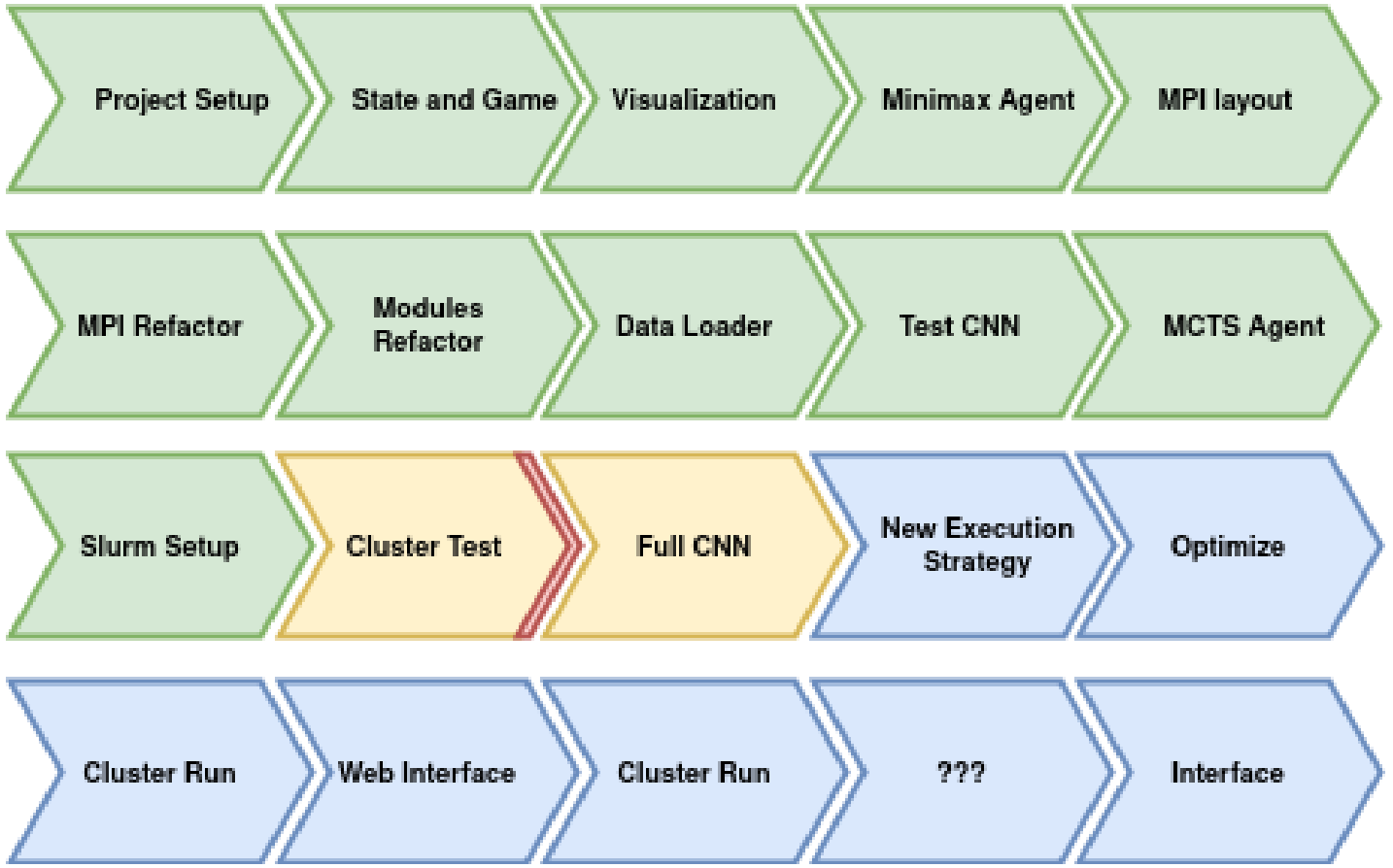
Slurm Usage

To get the project running using the Slurm Workload manager and conda first e

This necessitates a conda environment with the following requirements:



Milestones


Project Setup
State and Game
Visualization
Minimax Agent
MPI layout
MPI Refactor
Modules Refactor
Data Loader
Test CNN
MCTS Agent
Slurm Setup
Cluster Test
Full CNN
New Execution Strategy
Optimize
Cluster Run
Web Interface
Cluster Run
???
Interface



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